

Developing a Web Application in Django



Web applications have become critical components for the growth of a business, as they can handle a large number of different types of customers at the same time. Django, a Python based Web framework, enables you to create Web applications that are both secure and well-developed, in a very short period of time. This article discusses how to develop a basic Web application using Django.

Django is a free and open source Web framework developed and maintained by a non-profit organisation called the Django Software Foundation. Developing applications using Django makes things simple and easy.

In traditional Web application development, the frontend, the database and the part which joins the two are all developed separately. Django, however, takes care of the database connectivity as well as the connection between the frontend and the backend. A server like Xampp was earlier used to host and check the application. Django has an inbuilt server which helps to deploy and test the app.

In this article we will discuss how to develop a basic Web application using Django. We will develop a polling application where people can come and vote on different topics; so let us get started.

You can install Django using the following command:

```
$ Pip3 install django
```

To start a Django project, we use the following command:

```
$django-admin startproject poller
```

```
jishnusaarav@jishnusaarav-IdeaPad-L340-15IRH-Gaming:~$ django-admin startproject pollingpro
```

Figure 1: Files generated on creating a new project

Now run:

```
$Python3 manage.py runserver
```

If you go to port number 8000 on your localhost by typing `127.0.0.1:8000` in your browser, you will find your Web application running (Figure 2).

Once the server is set up and running, we can create our first app. Run:

```
$Python3 manage.py startapp poll
```

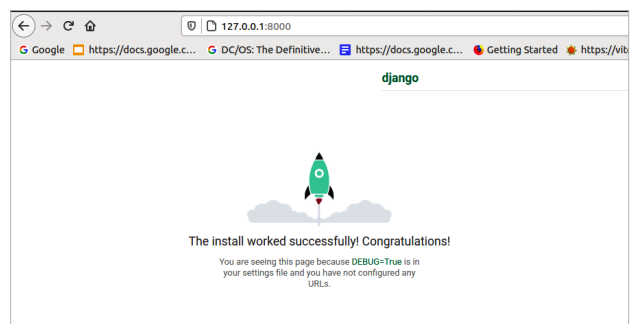


Figure 2: New project

In Django, every module is called an app. To develop each module, we have to create a new app for it using the above command. Now create a folder named *templates* in the *poller* folder. In the different files that are formed, we add the code as given below:

```
INSTALLED_APPS = [
    'polls.apps.PollsConfig',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'polls',
]
```

Basically, we need to include all the newly created apps in the 'installed apps' section, where there are already a few pre-created apps. What we do is to include our newly created app in this file every time an app is created.

```
poller/urls.py
```

```
urlpatterns = [
    path('polls/', include('polls.urls')),
    path('admin/', admin.site.urls),
]
```